online-learning-lean: Formal Proofs of FTRL Regret Bounds in Lean 4

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Abstract

`online-learning-lean` is a Lean 4 / Mathlib library formalising Follow-the-Regularised-Leader for online convex optimisation in a real inner product space. The development defines subgradient conditions, Bregman divergence, the FTRL setup, linearised regret, per-step stability and approximation bounds, a Bregman regret theorem, and the standard `O(sqrt K)` regret rate for bounded subgradients. The proof method is the classical Bregman three-point identity plus telescoping. The formal development is machine-checked in Lean 4 with zero `sorry`, zero `admit`, and standard Lean/Mathlib axioms only.

1. Introduction

Online convex optimisation models repeated decisions against a sequence of convex losses. At round `k`, the learner plays `x_k`, observes a subgradient `g_k`, and is evaluated by regret against a fixed comparator `u`. Follow-the-Regularised-Leader stabilises this process by balancing cumulative linearised loss with a strongly convex regulariser.

The repository formalises the deterministic core of that analysis. It does not model an adversary or probability distribution. Instead, it assumes a finite horizon `K`, a feasible set, iterates, subgradients, a Bregman regulariser, and the first-order optimality condition for each FTRL step. This is enough to prove the regret guarantee that is normally used as the algebraic backbone of FTRL analyses.

2. Mathematical Setting

The type `E` is a real inner product space. `OnlineLearning/Defs.lean` defines

```
IsSubgradientAt f g x :=
  forall y, f y >= f x + <g, y - x>
```

and the Bregman divergence

```
D_phi(x, y) = phi x - phi y - <gradphi y, x - y>.
```

An `FTRLSetup` packages the feasible set `C`, iterates `x`, subgradients `g`, horizon `K`, learning rate `eta`, regulariser `phi`, gradient `gradphi`, feasibility hypotheses, strong convexity as a Bregman lower bound, and the FTRL optimality condition. Regret against `u` is

```
sum_{k<K} <g_k, x_k - u>.
```

3. Main Theorems

`Stability.lean` proves `bregmanDiv_self`, `FTRLSetup.bregman_nonneg`, and `bregman_three_point`. It then proves `regret_decomposition`, splitting regret into stability terms and approximation terms, and `ftrl_stability`:

```
eta * <g_k, x_k - x_{k+1}> - D_phi(x_{k+1}, x_k)
  <= eta^2 * ||g_k||^2 / 2.
```

`Convergence.lean` proves `approximation_bound` and combines it with stability in `combined_per_step`:

```
eta * <g_k, x_k - u>
  <= D_phi(u, x_k) - D_phi(u, x_{k+1})
     + eta^2 * ||g_k||^2 / 2.
```

After summing, `bregman_regret_bound` gives

```
regret(u) <= D_phi(u, x_0) / eta
             + eta / 2 * sum_{k<K} ||g_k||^2.
```

With `||g_k|| <= G`, `ftrl_regret_bound` gives the bounded-gradient version, and `ftrl_convergence` proves the optimized rate

```
regret(u) <= G * sqrt(2 * D0 * K).
```

4. Proof Sketch

The proof is the standard Bregman telescoping proof. The three-point identity rewrites differences of Bregman divergences so that the FTRL optimality condition can control the approximation term. Young's inequality and the strong convexity lower bound control the stability term arising from moving from `x_k` to `x_{k+1}`.

The combined per-step theorem has the exact telescoping shape: `D_phi(u, x_k) - D_phi(u, x_{k+1})` appears on the right. Summing over `k < K` cancels all intermediate divergences and leaves only the initial comparator divergence plus a sum of squared subgradient norms. The final theorem substitutes the learning rate `eta = sqrt(2 * D0 / (K * G^2))` to obtain the `O(sqrt K)` bound.

5. Relation to Sibling Libraries

`online-learning-lean` is closest to `mirror-descent-lean`, DOI `10.5281/zenodo.20475033`, because both use Bregman divergences and a three-point identity. `sgd-lean`, DOI `10.5281/zenodo.20475583`, studies bounded-noise first-order optimisation, while `subgradient-lean`, DOI `10.5281/zenodo.20475946`, treats nonsmooth deterministic optimisation. `frank-wolfe-lean`, DOI

`10.5281/zenodo.20478157`, is another first-order method whose proof also reduces to a one-step inequality and a global rate.

6. Conclusion

`online-learning-lean` gives a small, importable Lean 4 proof of the FTRL regret theorem under explicit Bregman and bounded-subgradient hypotheses. Future work could instantiate the abstract regulariser with entropy or Euclidean squared norm, connect the subgradient interface to concrete convex losses, and develop executable online-learning examples on finite decision sets.

References

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